

# Lecture 1

## Vector Spaces and Subspaces

Unless stated otherwise,  $\mathbb{F}$  denotes either  $\mathbb{R}$  or  $\mathbb{C}$  and core problems are finite dimensional. Displayed function and sequence spaces are examples or bridges. Tags in parentheses indicate the flavour of a problem; a ★ marks a harder one.

### Core tutorial route

Work **1.1, 1.6, 1.8, 1.10, 1.18, 1.19, 1.24, and 1.25** in that order (about 65–80 minutes before discussion). This eight-problem route checks calculation, an axiom diagnosis, polynomial/sequence/function/matrix examples, scalar-field dependence, and both computational and explanatory direct-sum work.

**Additional practice:** 1.2–1.5, 1.7, 1.9, 1.11, 1.13–1.17, 1.21–1.23, 1.27. **Bridge:** 1.12 (ODE solution spaces). **Extension:** 1.20 and 1.26.

### Warm-up and axioms

**Exercise 1.1** (computational) In  $\mathbb{C}^2$ , let  $u = (1 + i, 2)$  and  $w = (3, -i)$ . Compute  $u + w$ , the vector  $(2 - i)u$ , and the additive inverse  $-u$ .

Answer.  $u + w = (4 + i, 2 - i)$ ;  $(2 - i)u = (3 + i, 4 - 2i)$ ;  $-u = (-1 - i, -2)$ .

**Exercise 1.2** (computational) In  $\mathbb{R}^3$ , let  $u = (2, -1, 3)$  and  $w = (0, 4, -2)$ . Compute  $u + w$ ,  $3u$ , and  $2u - w$ .

Answer.  $u + w = (2, 3, 1)$ ;  $3u = (6, -3, 9)$ ;  $2u - w = (4, -6, 8)$ .

**Exercise 1.3** (computational) In  $\mathcal{P}_3(\mathbb{R})$ , let  $p = 2 - t + 3t^2$  and  $q = 1 + 4t - t^2 + t^3$ . Compute  $p + q$ ,  $3p$ , and  $2p - q$ .

Answer.  $p + q = 3 + 3t + 2t^2 + t^3$ ;  $3p = 6 - 3t + 9t^2$ ;  $2p - q = 3 - 6t + 7t^2 - t^3$ .

**Exercise 1.4** (computational) In  $M_{2 \times 2}(\mathbb{R})$ , let  $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$  and  $B = \begin{pmatrix} 3 & -1 \\ 4 & 2 \end{pmatrix}$ . Compute  $A + B$ ,  $2A$ , and  $A - 2B$ .

Answer.  $A + B = \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}$ ;  $2A = \begin{pmatrix} 2 & 4 \\ 0 & -2 \end{pmatrix}$ ;  $A - 2B = \begin{pmatrix} -5 & 4 \\ -8 & -5 \end{pmatrix}$ .

**Exercise 1.5** (proof) Using only the vector space axioms, prove that  $av = 0$  (with  $a \in \mathbb{F}$ ,  $v \in V$ ) implies  $a = 0$  or  $v = 0$ .

Hint. Suppose  $a \neq 0$ . The field then hands you  $a^{-1}$ ; multiply the equation by it and read off  $v$ .

**Exercise 1.6** (proof) Decide, with proof, whether  $\mathbb{R}^2$  with the *usual* addition but the scalar multiplication  $a \odot (x, y) := (ax, 0)$  is a vector space over  $\mathbb{R}$ .

Hint. One axiom is enough to settle it — test the scalar 1 on a vector whose second coordinate is non-zero.

**Exercise 1.7** (proof) Prove the cancellation law in a vector space: if  $u, v, w \in V$  satisfy  $u + w = v + w$ , then  $u = v$ . Use only the axioms.

Hint.  $w$  has an additive inverse. Add it to both sides and let associativity do the rest.

## Recognising vector spaces

**Exercise 1.8** (computational) Which of the following are vector spaces under the natural operations?

1. the polynomials in  $\mathcal{P}(\mathbb{R})$  of degree *exactly* 3;
2. the polynomials  $p \in \mathcal{P}_3(\mathbb{R})$  with  $p(1) = 0$ ;
3. the real sequences  $(x_n)$  with  $x_n \rightarrow 0$ .

**Answer.** (1) No (not closed under +; also lacks 0). (2) Yes. (3) Yes.

**Exercise 1.9** (computational) Which of the following subsets of  $\mathbb{R}^2$  are vector spaces under the usual operations?

1.  $\{(x, y) : 2x - 3y = 0\}$ ;
2.  $\{(x, y) : x \geq 0\}$ ;
3.  $\{(x, y) : xy = 0\}$ .

**Answer.** (1) Yes. (2) No (not closed under scaling by  $-1$ ). (3) No (not closed under +).

**Exercise 1.10** (computational) Which of the following subsets of  $\mathbb{R}^{\mathbb{R}}$  (the real functions of one real variable) are vector spaces?

1.  $\{f : f(3) = 0\}$ ;
2.  $\{f : f(3) = 1\}$ ;
3.  $\{f : f \text{ is bounded}\}$ .

**Answer.** (1) Yes. (2) No (zero function fails  $f(3) = 1$ ). (3) Yes.

**Exercise 1.11** (computational) Which of the following subsets of the space of real sequences  $(x_n)$  are vector spaces?

1.  $\{(x_n) : x_1 = x_2\}$ ;
2.  $\{(x_n) : \text{only finitely many } x_n \text{ are nonzero}\}$ ;
3.  $\{(x_n) : x_n \rightarrow 5\}$ .

**Answer.** (1) Yes. (2) Yes. (3) No (zero sequence converges to  $0 \neq 5$ ).

**Exercise 1.12** (proof) Inside the vector space of twice-differentiable functions  $y : \mathbb{R} \rightarrow \mathbb{R}$ , show that the solutions of  $\ddot{y} + 5\dot{y} + 6y = 0$  form a vector space, but the solutions of  $\ddot{y} + 5\dot{y} + 6y = t$  do not.

**Hint.** Differentiation is linear, so check closure for the homogeneous equation. For the inhomogeneous one, ask whether the zero function is a solution.

**Exercise 1.13** (proof) Prove that  $\{p \in \mathcal{P}(\mathbb{R}) : p(1) = p(2)\}$  is a subspace of  $\mathcal{P}(\mathbb{R})$ , but that  $\{p \in \mathcal{P}(\mathbb{R}) : p(1)p(2) = 0\}$  is not.

**Hint.**  $p(1) = p(2)$  is a homogeneous linear condition on  $p$ ;  $p(1)p(2) = 0$  is not linear at all. For the second set, find two polynomials in it whose sum is not.

## Subspaces

**Exercise 1.14** (computational) For each subset of  $\mathbb{F}^3$ , decide whether it is a subspace.

1.  $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 0\}$ ;

2.  $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 4\}$ ;
3.  $\{(x_1, x_2, x_3) : x_1x_2x_3 = 0\}$ .

**Answer.** (1) Subspace. (2) Not a subspace (0 fails the condition). (3) Not a subspace (not closed under addition).

**Exercise 1.15** (computational) For each subset of  $\mathbb{R}^3$ , decide whether it is a subspace.

1.  $\{(x, y, z) : 2x - y = 0 \text{ and } z = 0\}$ ;
2.  $\{(x, y, z) : x^2 = y^2\}$ ;
3.  $\{(t, 2t, 3t) : t \in \mathbb{R}\}$ .

**Answer.** (1) Subspace. (2) Not a subspace. (3) Subspace.

**Exercise 1.16** (computational) For each subset of  $\mathbb{R}^4$ , decide whether it is a subspace.

1.  $\{(x_1, x_2, x_3, x_4) : x_1 - x_2 + x_3 - x_4 = 0\}$ ;
2.  $\{(x_1, x_2, x_3, x_4) : x_1 = x_4\}$ ;
3.  $\{(x_1, x_2, x_3, x_4) : x_1 + x_2 = 1\}$ .

**Answer.** (1) Subspace. (2) Subspace. (3) Not a subspace (0 fails the condition).

**Exercise 1.17** (computational) Decide whether each set is a subspace of the indicated space.

1.  $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$  in  $\mathcal{P}_3(\mathbb{R})$ ;
2.  $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 1\}$  in  $\mathcal{P}_3(\mathbb{R})$ ;
3.  $\{A \in M_{2 \times 2}(\mathbb{R}) : a_{11} = a_{22}\}$  in  $M_{2 \times 2}(\mathbb{R})$ .

**Answer.** (1) Subspace. (2) Not a subspace (0 fails). (3) Subspace.

**Exercise 1.18** (proof) Prove that the trace-zero matrices  $\{A \in M_{n \times n}(\mathbb{F}) : \text{tr } A = 0\}$  form a subspace of  $M_{n \times n}(\mathbb{F})$ , and that the invertible matrices do *not*.

**Hint.** The trace is linear, which settles the first set at once. For the invertible matrices, look at the zero matrix — or at  $I + (-I)$ .

**Exercise 1.19** (★) Is  $\mathbb{R}^2$  a subspace of the complex vector space  $\mathbb{C}^2$ ? Is it a subspace of  $\mathbb{C}^2$  regarded as a *real* vector space?

**Hint.** The deciding operation is multiplication by the complex scalar  $i$ : check whether  $i \cdot (1, 0)$  stays in  $\mathbb{R}^2$ .

**Exercise 1.20** (proof) Let  $U$  and  $W$  be subspaces of a vector space  $V$ . Prove that  $U \cup W$  is a subspace of  $V$  if and only if  $U \subseteq W$  or  $W \subseteq U$ .

**Hint.** One direction is immediate. For the other, argue by contradiction: if neither contains the other, pick  $u \in U \setminus W$  and  $w \in W \setminus U$  and examine  $u + w$ .

## Sums and direct sums

**Exercise 1.21** (computational) In  $M_{2 \times 2}(\mathbb{R})$ , write  $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$  as the sum of a symmetric and an antisymmetric matrix.

**Answer.**  $A = \begin{pmatrix} 4 & 4 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}$ .

**Exercise 1.22** (computational) In  $M_{2 \times 2}(\mathbb{R})$ , write  $A = \begin{pmatrix} 3 & 5 \\ 1 & 7 \end{pmatrix}$  as the sum of a symmetric and an antisymmetric matrix.

**Answer.**  $A = \begin{pmatrix} 3 & 3 \\ 3 & 7 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ .

**Exercise 1.23** (computational) In  $M_{3 \times 3}(\mathbb{R})$ , write  $A = \begin{pmatrix} 2 & 4 & 0 \\ 0 & 2 & 6 \\ 4 & -2 & 2 \end{pmatrix}$  as the sum of a symmetric and an antisymmetric matrix.

**Answer.**  $A = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 4 \\ 2 & -4 & 0 \end{pmatrix}$ .

**Exercise 1.24** (computational) In  $\mathbb{R}^3$ , for each pair of subspaces decide whether the sum  $U + W$  is direct.

1.  $U = \{(x, y, 0) : x, y \in \mathbb{R}\}$  and  $W = \{(0, 0, z) : z \in \mathbb{R}\}$ ;
2.  $U = \{(x, y, 0) : x, y \in \mathbb{R}\}$  and  $W = \{(u, 0, z) : u, z \in \mathbb{R}\}$ .

**Answer.** (1) Direct ( $U \cap W = \{0\}$ , and  $U \oplus W = \mathbb{R}^3$ ). (2) Not direct ( $U \cap W = \{(x, 0, 0) : x \in \mathbb{R}\} \neq \{0\}$ ).

**Exercise 1.25** (proof) Let  $U_e$  and  $U_o$  be the sets of even and of odd functions in  $\mathbb{R}^{\mathbb{R}}$  (so  $f(-x) = f(x)$  and  $g(-x) = -g(x)$ , respectively). Show that  $\mathbb{R}^{\mathbb{R}} = U_e \oplus U_o$ .

**Hint.** Every  $f$  splits as  $\frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$ ; check what each half is. For directness, ask what a function that is both even and odd can be.

**Exercise 1.26** (★) Give three subspaces  $V_1, V_2, V_3$  of  $\mathbb{R}^3$  such that  $V_i \cap V_j = \{0\}$  for every  $i \neq j$ , yet  $V_1 + V_2 + V_3$  is *not* direct. (This shows pairwise-trivial intersections are not enough for three or more subspaces.)

**Hint.** Pairwise-trivial intersection is weaker than directness — try three distinct lines lying in a single plane through the origin.

**Exercise 1.27** (proof) Let  $U$  and  $W$  be subspaces of a vector space  $V$ . Prove that  $U + W = W$  if and only if  $U \subseteq W$ .

**Hint.** Both steps are short. For the forward step (only if) write  $u = u + 0$ . The backward step (if) needs two inclusions:  $W \subseteq U + W$  follows from  $0 \in U$ , and  $U + W \subseteq W$  from closure of  $W$ .